10 · Did the site redesign help? (no control group) — interrupted time series (CausalPy)

July 9, 2026

1 10 · Did the site redesign help? (no control group) — interrupted time series (CausalPy)

> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. We redesigned the homepage on a known date, for **all** traffic at once — so, unlike the DiD store rollout (notebook 08), there is **no control group** to compare against. Conversion is up since the redesign. Is that real lift, or just the normal upward drift the site was already on? Keep the redesign or roll it back?

1.0.1 The idea: the past *is* the control group

When everyone is treated at the same moment, the only untreated comparison you have is **the treated unit's own history**. **Interrupted time series (ITS)** uses it: fit the pre-change series — its **level**, its **trend** (slope), and its **seasonality** (e.g. weekly cycles) — and then **extrapolate that fitted line forward** past the change date. That extrapolation is the **counterfactual**: "what conversion *would* have done if we'd changed nothing." The **gap** between actual conversion and this projection, after the change, is the estimated effect.

Because it's a forecast, the counterfactual comes with a **credible band that widens the further out we project** — an honest admission that "what would have happened" is less and less certain over time.

1.0.2 The assumption and its checks

ITS is the *weakest* of the quasi-experiments because it leans entirely on **no concurrent shock**: nothing *else* (a price change, a competitor move, a seasonality break) happened at the same date as the redesign, or ITS will blame the redesign for it. We stress this with a **placebo-in-time** (pretend the change happened earlier — the fake effect should be ~ 0), track the **cumulative** impact, and check **residual autocorrelation** (successive days being correlated, which — if ignored — makes the intervals look tighter than they should).

> **On real data.** ITS fits any "everyone changed at once on a known date" event: a site redesign, a pricing change, a policy launch, a PR crisis. You need a reasonably long, stable **pre-period** and a clean intervention date. A famous public example is the effect of anti-smoking laws on cigarette sales.

7-step contract.

1.1 2 · Simulate a ground truth

Daily conversion with a gentle upward trend and weekly seasonality. The redesign lands on **day 120** for everyone and adds a **true +3pp** step (~+30% relative on a ~10% base). The trend is the confounder (conversion was already drifting up); the step is the effect.

```
TRUE redesign effect = +3pp conversion from day 120

   t  day      sin7      cos7  conversion
t
0  0    0  0.000000  1.000000    0.097628
1  1    1  0.781831  0.623490    0.109552
2  2    2  0.974928 -0.222521    0.113692
3  3    3  0.433884 -0.900969    0.098956
4  4    4 -0.433884 -0.900969    0.100867
```

1.2 3 · Identify — extrapolate the pre-trend as the counterfactual

Fit the pre-period series $f(t)$ (level, slope, weekly seasonality) and extrapolate past the intervention as the no-redesign counterfactual $\hat{Y}_t(0)$; the effect is $Y_t - \hat{Y}_t(0)$ afterward. Being Bayesian, the counterfactual is a posterior-predictive band that **widens with horizon** — honest extrapolation. **Identification assumption (the weakest of the quasi-experimental set): no concurrent shock** at the redesign date (no competing campaign, price change, or seasonality break landing at the same time). Best when the change hit everyone at once and the pre-period is long and stable.

1.3 4 · Estimate — Bayesian interrupted time series

We fit the pre-period conversion series with a trend term (`t`) and weekly seasonality (`sin7`, `cos7` Fourier terms — a smooth way to encode the day-of-week cycle), then let the model extrapolate that fit past the redesign day. The **effect** is the average gap between actual conversion and the extrapolated no-redesign line over the post-period. The `formula` is literally the shape of the pre-trend we're projecting forward — get the seasonality wrong and the counterfactual is wrong, so it matters.

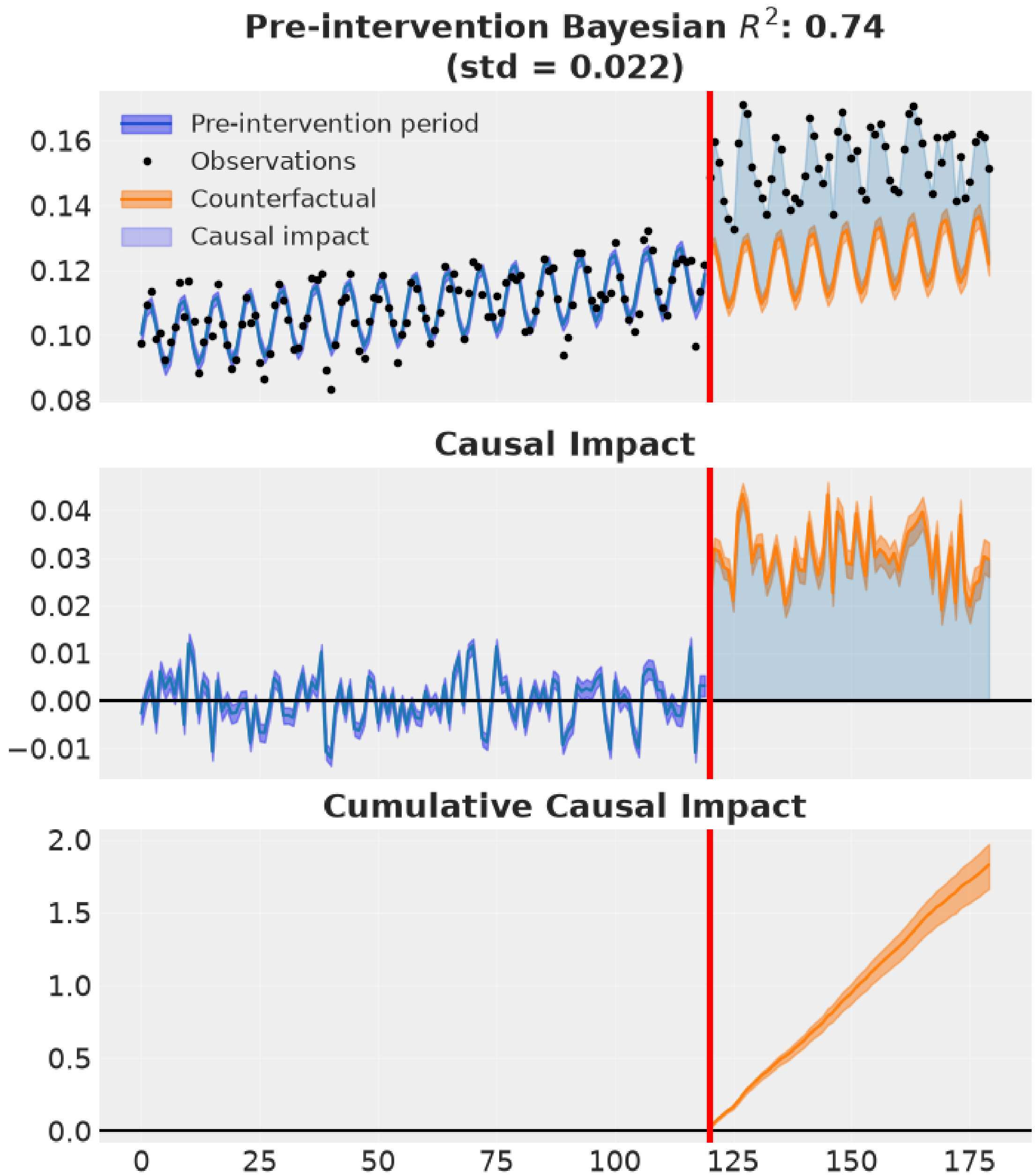
```
ITS lift +3.0pp (true +3pp)  ·  90% credible interval (CrI) [+2.8pp, +3.3pp]
ITS convergence: max r-hat 1.000 - min ESS 2918 - divergences 0
```

1.4 5 · Validate — counterfactual, cumulative impact, placebo, autocorrelation

Two figures, four checks:

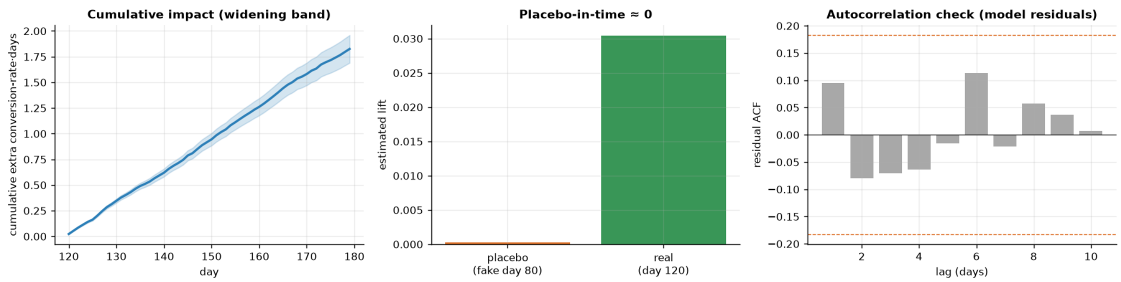
1. **Counterfactual** (CausalPy's own plot) — observed vs the extrapolated no-redesign path with its widening band, and the impact below.
2. **Cumulative impact** — total extra conversions accrued since launch, with uncertainty.
3. **Placebo-in-time** — pretend the redesign happened *before* it did; the fake effect should be ~0.
4. **Residual autocorrelation** — daily series are autocorrelated; if the model's residuals still are, the uncertainty is understated.

How to read the CausalPy fit figure. The header reports a **Bayesian R^2** — the share of the pre-period variance the fitted level+trend+seasonality model explains, computed from the posterior predictive (so it carries uncertainty rather than being a single point estimate). A high value (~**0.74**) means the pre-period model **tracks the observed series well**, which is what earns us the right to extrapolate that fit forward as the counterfactual. The **shaded bands** are posterior-predictive prediction intervals — around the fitted line before the redesign, and around the projected no-redesign path after it; they **widen with horizon**, the honest statement that the counterfactual grows less certain the further past the change date we project. (Note: the cumulative-impact panel shows up both in this figure and in the next one — the same quantity via two code paths, left as-is.)



```
recovered average step +3.0pp vs true +3pp
```

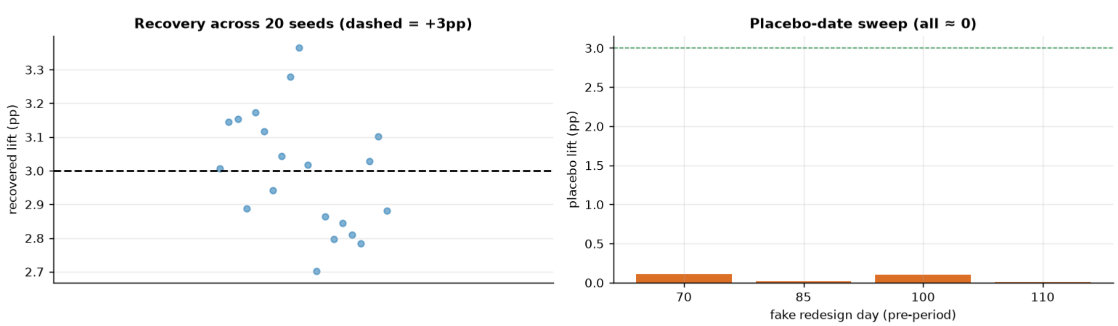
```
placebo-in-time +0.0% (~0, no spurious pre-effect)  ·  residual ACF: 0/10 lags
outside the +/-2/sqrtn band (lag-1 +0.09) -> residuals ~ white, so the model's
intervals are not understated.
```



How to read this. *Left* — the **cumulative** impact (total extra conversions accrued since the redesign) climbs steadily, and its band **fans out** the further we project — the honest statement that long-horizon totals are less certain. *Middle* — the key falsification: we re-ran the whole method with a **fake** redesign date back in the pre-period; the fake effect is ~ 0, so the method isn't inventing lift where none exists. If this bar had been large, our real estimate would be suspect. *Right* — the residual autocorrelation (ACF) of the **model's own residuals** (after fitting the same trend + weekly-seasonality terms the ITS uses): essentially every bar sits inside the +/-2/sqrtn band, so successive days are effectively uncorrelated and the model's intervals are not understated. (Computing this ACF on a crude rolling-mean "residual" instead would leave the weekly cycle in and fake an alarming lag-7 spike — a self-inflicted false alarm we avoid by residualising against the actual model.)

1.4.1 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test is whether the estimator is **centred on the truth over repeated samples** and whether its 90% interval **covers** the truth at the stated rate. We refit on many fresh samples (a small fast fit each) and check both.



```
ITS across 20 seeds: mean +3.00pp (true +3pp)  bias -0.00pp  sd 0.17pp  ·  90%
interval covers truth in 17/20 seeds.
Placebo-date sweep (pre-period fakes): day 70 +0.11pp, day 85 +0.02pp, day 100
+0.10pp, day 110 +0.01pp - all near zero, so the method doesn't manufacture
effects where none exist.
```

1.5 6 · Decide, in euros — keep or roll back?

A redesign lift is rarely permanent — **novelty fades**. Projecting the measured 60-day lift *flat* across a whole year (a naive `lift x visitors x value x 365`) contradicts this notebook's own "uncertainty grows with horizon" message. So we decide against an explicit **redesign cost** and sweep the **persistence**: the effect decays with a novelty half-life, and we discount future days accordingly (effective days = $\sum_t 0.5^{\,t/\text{half-life}}$). For each persistence scenario we report the annual value and **P(annual value > cost)**. The 0.9 go/no-go bar is the cookbook's convention — act only when the posterior is at least 90% sure the value clears the cost.

```
Undecayed lift: 243 extra conversions/day  ·  €9,731/day
Assumed amortized redesign cost: €600,000/year

 novelty half-life | eff. days | annual value (mean) | P(value>cost) | verdict
          no decay |       365 | €        3,551,767 |          1.00 | KEEP
           6-month |       196 | €        1,910,943 |          1.00 | KEEP
           3-month |       123 | €        1,192,075 |          1.00 | KEEP
           1-month |        44 | €          425,952 |          0.00 | roll back
/ renew

Break-even lift (no decay): 0.51pp of conversions covers the €600,000 cost; we
estimate 3.0pp.
P(redesign helped at all) >0.99 (4,000 draws) -> KEEP under any realistic
persistence; only a harsh 1-month novelty half-life turns it into a re-test.
```

1.6 7 · Caveats

- **The no-concurrent-shock assumption is the weak link.** A pricing change or competitor move at the redesign date gets attributed to the redesign. Vet the calendar around the intervention.
- **Uncertainty grows with horizon** — the band (and cumulative-impact band) widens; long-horizon claims are genuinely less certain.
- **A control group beats extrapolation.** If even a small hold-out or comparable untreated series exists, DiD (nb 08) or synthetic control (nb 07) are stronger. ITS is the last resort when *everyone* was treated.
- **Autocorrelation.** If residuals are autocorrelated (checked above), a naive model understates uncertainty; prefer trend/seasonality structure or explicit AR terms.